

Applications of Algebra (Part II) Homework

① If \$18,000 is deposited in a bank account paying a 2% annual interest rate compounded quarterly, how much money will be in the account in 3 years? Assume that there are no withdrawals or additional deposits.

② If \$1,090 is deposited in a bank account paying a 1.7% annual interest rate compounded semiannually, how much money will be in the account in 4 years? Assume that there are no withdrawals or additional deposits.

③ Money is deposited in an account paying a 1% annual interest rate compounded daily. After 8 years, the investment grows to \$5,562. How much money was initially deposited in the account? Assume that there were no additional deposits or withdrawals.

④ The bank gave Gary a \$2,000 personal loan, but 2 years have past since he received the money and he has not attempted to pay his debt. Consequently, the loan has grown to \$3,002. What annual interest rate did the bank charge if it was compounded monthly?

⑤ Jeff and Linda take out a mortgage on a house. The interest is compounded annually at a 4.7% annual rate. If they do not make payments for an initial period, and the debt grows from \$300,000 to \$389,412, how long was this initial period that they did not make payments?

⑥ If interest is compounded continuously for 6 years at a 1.8% rate, how much would a \$3,500 investment grow to?

⑦ If a 4% interest rate is compounded continuously for 5 years, and the initial investment is \$600, how much would the investment be worth?

⑧ If a \$20,000 investment grows to \$25,334 in 7 years, what interest rate was applied if it was compounded continuously?

⑨ If an investment grows from \$819 to \$1200 at a 14% annual interest rate compounded continuously, how much time passed during this growth?

Answers

- ① \$19,110.20
- ② \$1,166.36
- ③ \$5,134.38
- ④ 20.48%
- ⑤ 5.6795 years
- ⑥ \$3,899.17
- ⑦ \$732.84
- ⑧ 3.38%
- ⑨ 2.7285 years